

YITIAN WANG

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SUMMARY

Ph.D. scholar in Electrical Engineering specializing in thermal transport and lattice dynamics in functional oxides and solid-state ionic conductors. Experienced in crystal and thin-film growth, vacuum deposition, material characterization, and data-driven modeling using Python and MATLAB. Published 5 first-author works in leading journals and skilled at collaborating with national labs and cross-disciplinary device and materials teams. Additional teaching assistant experience during Ph.D. studies.

EDUCATION

University of California, Riverside, CA Ph.D. in Electrical and Computer Engineering	Mar. 2021 – Jun. 2026 (expected)
Columbia University in the City of New York, NY Master of Science, Material Science and Engineering	Aug. 2019 – Feb. 2021
University of California, Berkeley, CA Summer Session, Physics	Jul. 2017 – Aug. 2017
Beijing Normal University, CN Bachelor of Science, Physics	Sep. 2015 – Jun. 2019

SKILLS

Software	Python, MATLAB, R, SQL, LaTeX, COMSOL, SolidWorks, Linux
Analysis Tools	GSAS-II, Mantid, ImageJ, VESTA, Quantum ESPRESSO, VASP
Synthesis & Growth	Floating-Zone Crystal Growth, PVD / Magnetron Sputtering, CVD
Characterization	SEM, TEM, Scanning Tunneling Microscopy (STM), XRD, AFM, Inelastic Neutron Scattering, Raman, FTIR
Transport Properties	PPMS, DSC, TGA, Four-Probe Transport, Mechanical Strength Testing
Battery & Electrochem	Prototype Cell Assembly, Separator Fabrication, EIS, Galvanostatic Cycling, Thermal Abuse Testing
Methods	Design of Experiments (DOE), AI Agent Training

AWARDS & ACTIVITIES

- Dissertation Completion Fellowship Award, University of California, Riverside (2025)
- MRS 2023 Spring Meeting Highly Commended Student Talk Award
- ‘Jingshi’ Scholarship, Beijing Normal University (2016 – 2018)
- COMAP Mathematical Contest in Modeling — team member and team leader

RESEARCH EXPERIENCE

Graduate Research Assistant <i>University of California</i>	Jun. 2021 – Jun. 2025 <i>Riverside, CA</i>
• Pioneered lattice-dynamics studies on Li-ion solid-state electrolytes, extending work previously limited to Cu- and Ag-ion systems. • Published 5 first-author papers in journals including <i>PRX Energy</i> and <i>J. Mater. Chem. A</i>. • Built a floating-zone crystal growth workflow from scratch, producing centimeter-scale LLZTO single crystals that enabled phonon-resolved neutron scattering measurements previously infeasible on polycrystalline samples.	

- Identified *diffuson*-mediated thermal transport in Li-ion conductors, providing the first verification of this mechanism in Li-ion solid-state electrolytes.
- Developed a two-channel (phonon + *diffuson*) thermal-conductivity model in MATLAB that explained anomalous temperature dependence in Li-ion conductors where the traditional Debye model failed.

Experiment Lead

Oak Ridge National Laboratory

Jan. 2022 – Jul. 2023

Oak Ridge, TN

- Authored two successful beam time proposals and led inelastic neutron-scattering experiments on LLZTO single crystals using TAX (HFIR) and ARCS (SNS) at Oak Ridge National Laboratory (ORNL).
- Resolved the full phonon dispersion of a garnet solid electrolyte for the first time, providing experimental benchmarks for ab initio lattice-dynamics calculations.
- Built automated data-reduction pipelines in Mantid and Python, significantly reducing manual analysis time.
- Managed on-site experimental decisions and coordinated across ~7 research groups spanning UCR, UT Austin, ORNL, ETH Zürich, and TU Wien.

Student Researcher

Columbia University

Sep. 2019 – Jan. 2021

New York, NY

- Engineered composite Li-ion battery separators (γ -C₃N₄/PVDF) via solution processing, targeting improved electrochemical reaction kinetics.
- Assembled and tested CR2032 full cells over 100+ galvanostatic cycles, characterizing cycling performance and interfacial stability via EIS.
- Applied first-principles DFT (Quantum ESPRESSO) to model defect stability in olivine cathode structures, complementing experimental work with computational materials design.

Student Researcher

Beijing Normal University

Sep. 2018 – May 2019

Beijing, CN

- Designed a uniaxial-strain apparatus in SolidWorks exploiting differential thermal contraction for controlled detwinning of iron-based superconductors below 140 K.
- Built a Monte Carlo simulation in MATLAB to model atomic movement of metal atoms inside multi-wall carbon nanotubes, reproducing experimental observations.

TEACHING EXPERIENCE

Graduate Teaching Assistant

University of California, Riverside

Technology in the Premodern World (ENGR 108)	Reader	Fall 2025	
Circuits and Electronics Lab (EE 005)	Lab Instructor	Spring 2025	7.0/7.0 (96th %ile)
Technology in the Premodern World (ENGR 108)	TA / Reader	Spring 2025	5.7/7.0 (83rd %ile)
Linear Methods for Engr. Analysis – MATLAB (EE 020B)	Discussion TA	Winter 2023	6.5/7.0 (81st %ile)
Magnetic Materials (EE 139)	Discussion TA	Fall 2022	6.5/7.0 (77th %ile)

- Served 280+ students across 7 sections over 3 academic years.
- Achieved perfect 7.0/7.0 evaluation (96th percentile campus-wide) in EE 005, with consistently above-department averages across all courses.
- Selected student feedback: “singlehandedly one of the most effective teachers on this campus” and “top tier educator” (EE 005); “the backbone of the entire class” (EE 020B); “always helpful... feedback was helpful in creating a well organized presentation” (EE 139).

INTERNSHIP EXPERIENCE

Administrative Assistant, Physics Department

Beijing Normal University

Sep. 2018 – Jan. 2019

Beijing, CN

- Managed archival of 3 years of departmental paperwork and organized the academic calendar for 4 course sections.

Student Intern

Shanghai East China Computer Corp.

Jul. 2016 – Aug. 2016

Shanghai, CN

- Gained hands-on exposure to IT industry workflows, strengthening teamwork and cross-functional communication skills.

SELECTED PUBLICATIONS

1. **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen. “Low thermal conductivity and lattice anharmonicity of NaSICON-type solid electrolyte $\text{Na}_3\text{Zr}_2\text{Si}_2\text{PO}_{12}$.” *Tungsten*, 2025. [\[Link\]](#)
2. **Y. Wang**, Q. Jia, S. Li, L. Shi, Y. Li, X. Chen. “Glass-like thermal transport in polycrystalline perovskite Li-ion conductor $\text{Li}_{3/8}\text{Sr}_{7/16}\text{Hf}_{1/4}\text{Ta}_{3/4}\text{O}_3$.” *Chem. Commun.*, 2025. [\[Link\]](#) [\[Front Cover\]](#)
3. **Y. Wang**, Y. Su, J. Carrete, H. Zhang, N. Wu, *et al.* “Origin of intrinsically low thermal conductivity in a garnet-type solid electrolyte.” *PRX Energy*, 2025 (OA fee waived). [\[Link\]](#)
4. S. Li, S. Guo, **Y. Wang**, H. Li, Y. Xu, V. Carta, J. Zhou, X. Chen. “Enhanced magnon thermal transport in yttrium-doped spin ladder compounds $\text{Sr}_{14-x}\text{Y}_x\text{Cu}_{24}\text{O}_{41}$.” *J. Appl. Phys.*, 2024. [\[Link\]](#)
5. **Y. Wang**, S. Li, N. Wu, Q. Jia, T. Hoke, L. Shi, Y. Li, X. Chen. “Thermal properties and lattice anharmonicity of Li-ion conducting garnet solid electrolyte $\text{Li}_{6.5}\text{La}_3\text{Zr}_{1.5}\text{Ta}_{0.5}\text{O}_{12}$.” *J. Mater. Chem. A*, 2024. [\[Link\]](#)
6. S. Guo, H. Li, X. Bai, **Y. Wang**, S. Li, R. E. Dunin-Borkowski, J. Zhou, X. Chen. “Size-dependent magnon thermal transport in a nanostructured quantum magnet.” *Cell Rep. Phys. Sci.*, 2024. [\[Link\]](#)
7. Y. Xu, Z. Barani, P. Xiao, S. Sudhindra, **Y. Wang**, *et al.* “Crystal structure and thermoelectric properties of layered van der Waals semimetal ZrTiSe_4 .” *Chem. Mater.*, 2022. [\[Link\]](#)
8. **Y. Wang**, X. Chen. “Single crystal growth and electrochemical studies of garnet-type fast Li-ion conductors.” *Tungsten*, 2022. [\[Link\]](#)